

OFFICIAL SLIDE DECK

PROJECT SENTINEL

AI-Powered Intelligent CCTV Investigation & Multi-Agent Surveillance Platform

"Turning existing CCTV infrastructure into a coordinated intelligence network."

Scalable AI architecture designed for 80,000+ state CCTV nodes. Built to run on Gujarat's existing camera infrastructure — zero hardware replacement.

Gujarat Has Cameras. The Challenge Is Connecting Their Intelligence.

A large CCTV network produces enormous volumes of continuous video. During a critical vehicle incident or crime investigation, officers face time-critical questions:

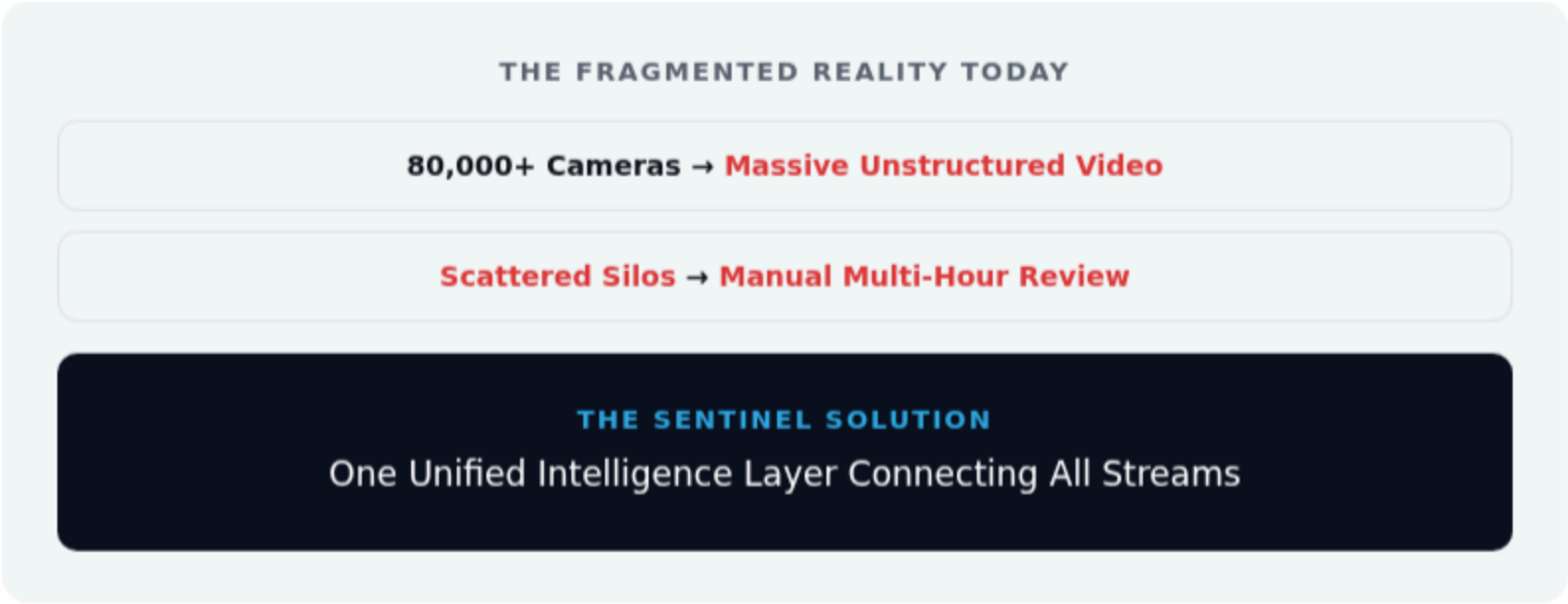
- Which cameras saw it?

Exact timestamp of sighting?

Where did it move next?

Matches official records?

Key Insight: The problem is not watching cameras — it is rapidly querying data across thousands of isolated feeds.



Sentinel-AI: A High-Performance, Microservices-Based Hybrid Architecture

Designed to integrate 80,000+ CCTV nodes without replacing existing hardware or VMS infrastructure.

01 · Video Ingestion & WebRTC Streaming

RTSP feeds over TCP via OpenCV, transcoded to WebRTC via MediaMTX for ultra-low-latency browser playback.

02 · Dynamic Geo-Fenced AI Micro-Agents

Spawned within a 5km radius of a reported incident using PostGIS spatial queries — not continuous 24/7 scanning.

03 · Dual-Mode Analytics

Targeted single-vehicle deep search alongside automated batch-logging for real-time watchlist cross-referencing.

04 · Database Partitioning & Redis Caching

State-code sharding (GJ01, GJ04...) with in-memory Redis lookup for target sub-millisecond vehicle queries.

05 · Decoupled Web Architecture

Frontend/APIs kept fully separate from heavy AI compute, protecting dashboard uptime under heavy AI load.

How Sentinel Works: Ingestion & Streaming

Layer 1 & 2 — Existing Cameras & RTSP Ingestion

Connects to 80,000+ CCTV nodes without replacing hardware.

Layer 3 — WebRTC Browser Streaming

Ultra low-latency video streaming directly to command center browsers.

Layer 4 — AI Analytics / Core Computer Vision Pipeline

Detects vehicles, number plates, occupants, and unusual movement patterns.

How Sentinel Works: Intelligence & Command

Layer 5 — Innovation: Dynamic AI Micro-Agents (5 KM Geo-Fence)

Spawns targeted lightweight AI workers only around active incident zones.

Layer 6 — Intelligence: Sharded DB & Fast Redis Cache

Cross-references plate sightings with government records at target sub-millisecond speed.

Layer 7 & 8 — Command & Human: GIS Dashboard & Human Decision Loop

AI highlights intelligence → Human officer verifies → Immediate police action.

Instead of Searching the Whole City, Search Where It Matters

An incident — a watchlist hit, a flagged plate, or a manual investigator query — triggers agent spawning. Using PostGIS spatial queries against camera GPS metadata, Sentinel identifies all cameras within a 5 KM radius of the trigger location and spins up lightweight detection workers scoped only to those feeds. Agents release back to standby after an idle window with no new activity.

CONVENTIONAL APPROACH

Full-Network Continuous Processing

Runs heavy AI models on all 80,000+ camera feeds simultaneously, 24/7.

Massive server cost, excessive power consumption, high latency.

SENTINEL AGENTIC APPROACH

Dynamic 5 KM Geo-Fenced AI Micro-Agents

When an incident triggers, Sentinel spawns lightweight AI agents assigned specifically to nearby cameras within a 5 KM radius.

Simple Analogy: Deploying investigators to the scene instead of searching the entire state.

Dual-Mode AI Analytics: Two Ways of Looking at Vehicles

Both modes operate concurrently on the unified Sentinel ingestion backbone.

MODE 1 • TARGETED INVESTIGATION

Triggered for a specific target vehicle (e.g., license GJ01AB9999). Full computer-vision pipeline — occupant detection, driver crops, deep tracking — activates only within an active 5 KM incident zone.

- Search specific camera feeds
- Reconstruct movement history
- Extract driver/occupant crops

Use Case: High-priority active crime response.

MODE 2 • CONTINUOUS VEHICLE LOGGING

Background recording of passing vehicle observations, using a lightweight, plate-only detector — cheaper than Mode 1's full pipeline, so the 90%+ compute-savings target holds even while every camera logs continuously.

- > Index plate + time + junction
- > Cross-reference with watchlists
- > Instantly match suspicious plates

Use Case: Watchlist alerts & audit trail history.

Dynamic Geo-Fenced AI Micro-Agents

Spawns lightweight AI workers within a 5km radius of a reported incident, reducing statewide GPU/CPU overhead by 90%+ (Design Target).

Sharded & Redis-Cached Lookups

State-code partitioning (GJ01, GJ04...) with in-memory Redis caching, designed to handle 400,000+ plate queries/sec (Design Target).

Occupant & Driver Snapshots

High-resolution driver/occupant facial crops alongside plates for enhanced suspect identification.

Zero-Hardware-Overhaul Integration

Direct ingestion of existing RTSP-over-TCP feeds from multi-vendor VMS/DVR/NVR — no camera replacement.

Dual-Mode Processing

Targeted single-vehicle deep tracking alongside automated batch-logging and real-time watchlist cross-referencing.

10-Second Evidence Clipping

Auto-cuts and archives a 10-second clip (5s pre / 5s post detection) to cloud/MinIO storage for instant legal proof.

WebRTC GIS Dashboard

RTSP → WebRTC via MediaMTX, interactive Leaflet.js mapping, animated trajectories, instant audio alerts.

From Number Plate to Complete Investigation Trail

- 1

Plate Recognition

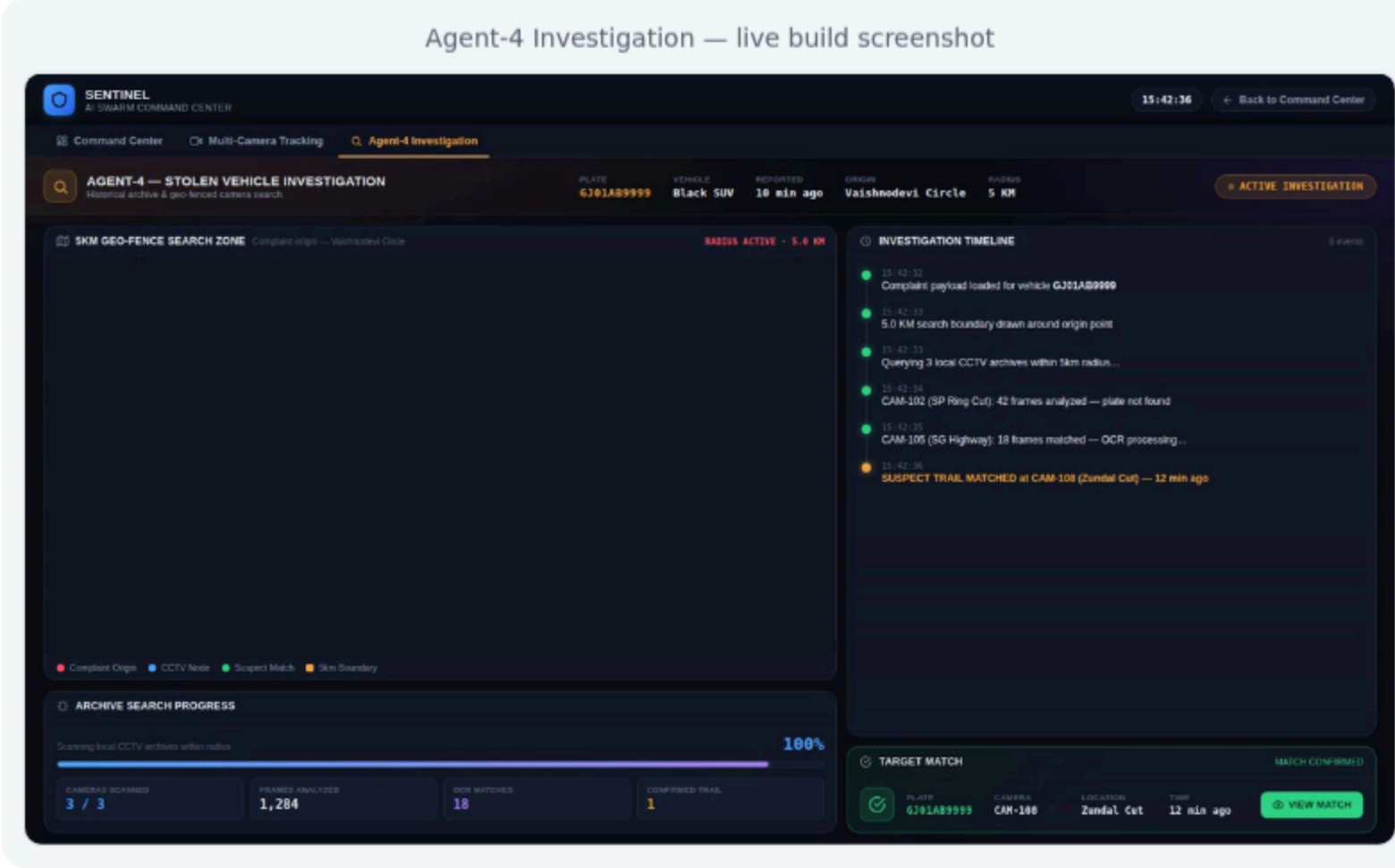
AI extracts plate text + camera timestamp.
- 2

Historical Cross-Reference

Queries past sightings across all state nodes.
- 3

Trajectory Reconstruction

Maps vehicle movement step-by-step on GIS.



Outcome: Investigators view a clear chronological map of vehicle sightings instead of watching raw video logs.

How Sentinel Handles Millions of Fast Vehicle Queries

State-Code Sharding

Database Sharding: Splitting a giant database into smaller regional sections (e.g., GJ01, GJ04, GJXX) so queries run faster.

Partition → GJ-01 (Ahmedabad) | GJ-04 (Bhavnagar)

Redis In-Memory Cache

Redis Cache: Very fast temporary memory keeping active vehicle watchlists immediately accessible.

RAM Cache Lookup Time: Target < 0.5 ms

Query Throughput Target

400,000+

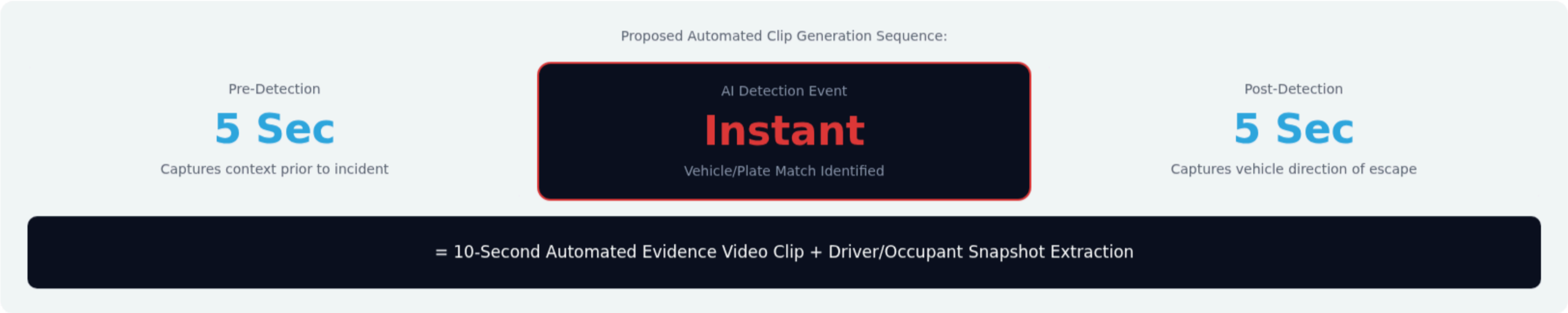
Plate Queries / Sec

Latency Target

<0.5ms

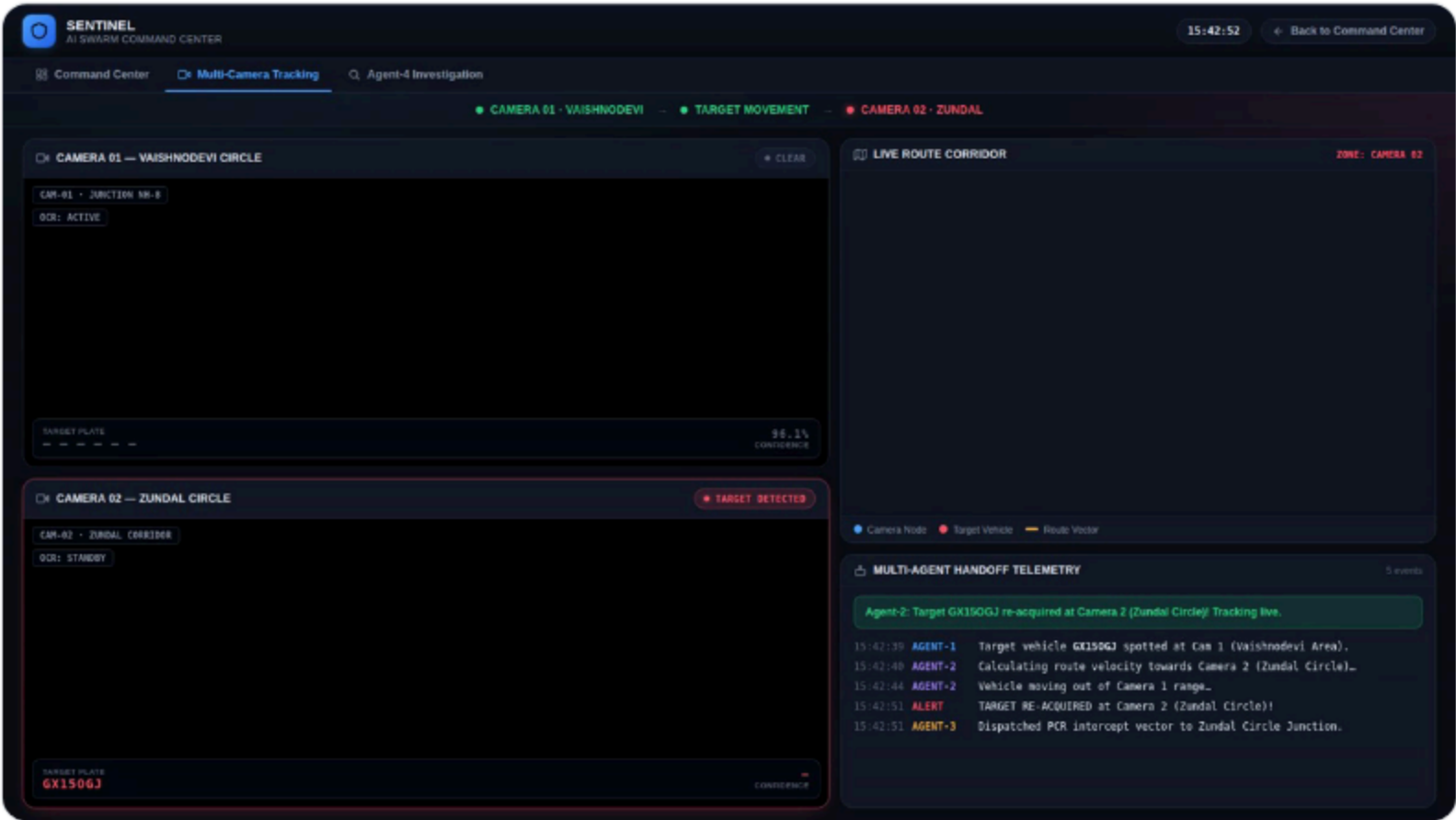
Vehicle Lookup

Automated Evidence Generation: 10-Second Smart Clips



Note: Proposed automated feature designed to assist investigators. Every clip is hash-logged and stored read-only, preserving chain-of-custody integrity — evidence packaging remains subject to standard legal verification protocols.

Live GIS Command Dashboard: One Map, Unified Intelligence



Geospatial Mapping

Live camera locations & active 5 KM incident zones.

WebRTC Live Video

Low-latency live stream popups on click.

Automated Alerts

Instant notifications when target vehicle is spotted.

Multi-Agency Command: Shared Platform Architecture



Health & Panchayat feeds are included in the shared camera backbone under non-investigative, view-limited access.

90%+ Infrastructure Cost Savings (Design Target)

Event-triggered geo-fenced agents vs. continuous statewide scanning cuts GPU compute, server memory, and cloud bandwidth expenses.

High-Throughput Database Stability

Sharding + Redis caching eliminates I/O bottlenecks, reliably handling 400,000+ plate queries/sec at peak load.

Unified Multi-Agency Command

One WebRTC GIS dashboard for Police, RTO, and Municipal teams, with layer toggling and real-time alerts.

Sub-3-Second Incident Response (Design Target)

Reduces vehicle trajectory reconstruction from hours of manual CCTV playback to under 3 seconds.

Instant Legal Evidence Generation

Automated 10-second proof clipping plus high-res occupant/driver crops removes manual DVR extraction delays.

High Availability & Network Resilience

Containerized microservices target 100% dashboard uptime, backed by exponential-backoff auto-reconnection.

Role-Based Access Control

Each department — Police, RTO, Municipal — sees only what it's authorized to see.

Full Audit Logging

Every officer query is logged and reviewable, preventing misuse.

Evidentiary Integrity

Auto-generated clips are hash-logged and read-only, preserving chain-of-custody for court use.

Encryption

Data encrypted in transit (TLS) and at rest for all stored evidence and query logs.

Data Retention Policy — PLACEHOLDER

[Team to define retention windows — e.g. non-flagged Mode 2 logs retained 30/90 days; flagged/watchlist matches retained per case lifecycle.]

Resilience: Built for a Network That Cannot Stop

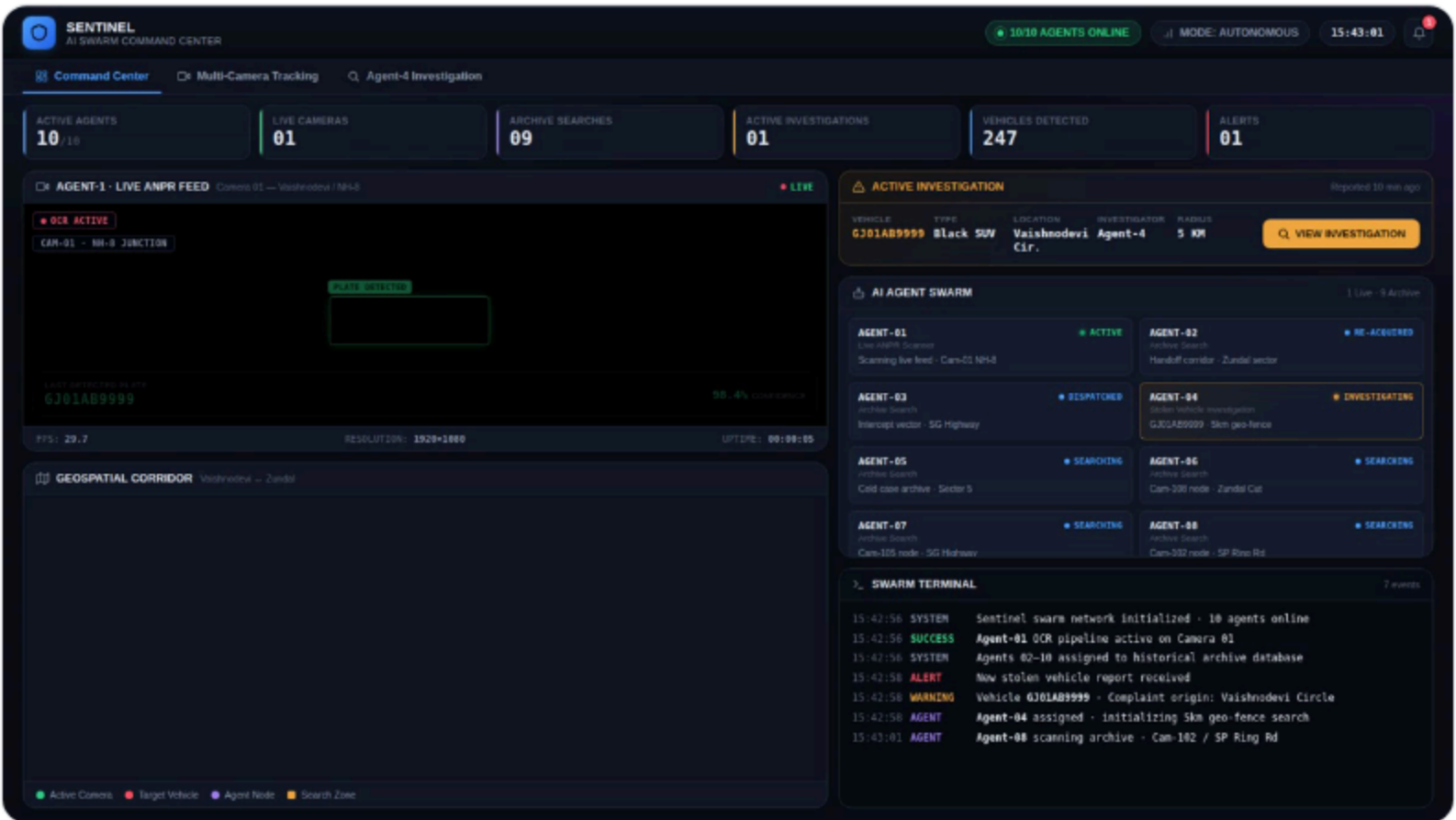
Containerized Microservices

Microservices: Packaging system functions into independent workers. If heavy AI processing spikes, web streaming and database lookup remain unaffected.

Network Auto-Reconnection

If network connection drops at a regional camera node, local buffering holds footage and automatically reconnects without human intervention.

What We've Actually Built



AI Swarm Command Center

Live ANPR feed panel, real-time KPI bar (active agents, archive searches, alerts), geospatial corridor map, 10-agent swarm grid, and a scrolling swarm terminal — all functioning and wired to live state.

Agent-4 Investigation View

5km geo-fence search zone, archive search progress bar with live stats, investigation timeline, and target-match reveal panel.

Multi-Camera Tracking

Dual camera feed cards with live handoff, route corridor map, and multi-agent handoff telemetry log.

Still needed before submission

[INSERT: number of real cameras actually connected · real detection count from live testing · any genuinely measured latency numbers, to sit alongside the design targets elsewhere in this deck.]

The Team Behind Sentinel

Harsha Lohana

TECH LEAD & BUILDER

- Structure of the product
- Functionality and other work
- Project coordination

Jignesh Thakkar

BUILDER AND PRESENTER

- Product look and feel
- Features and use cases
- Branding and presentation
- Research and pitching

Nirav Prajapati

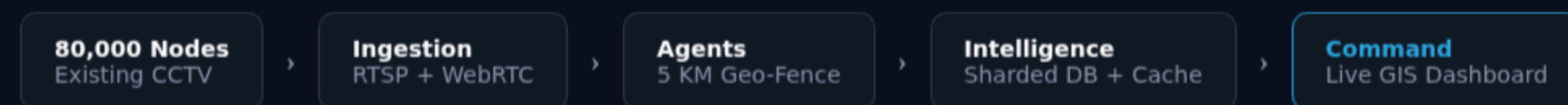
DEVELOPER AND BUILDER

- Development support
- Logical thinking
- Research work

THE FINAL VISION · GUJARAT POLICE INNOVATION HACKATHON 2026

FROM 80,000+ CAMERAS TO ONE INTELLIGENT PICTURE

Project Sentinel is designed to transform existing CCTV infrastructure into a scalable, intelligent investigation network — without replacing a single camera.



PROJECT SENTINEL

Gujarat Police Innovation Hackathon 2026

PROTECT WHAT MATTERS.